

AI-ASSISTED INCIDENT RESPONSE FOR LOG ANALYSIS AND BRUTE-FORCE ATTACK DETECTION

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ABSTRACT

The rising number of cybersecurity threats has created a challenging situation for security analysts to deal with the increasing complexity of cybersecurity threats using conventional manual incident response techniques. With the increasing amount of log data being produced by organizations, it is critical to analyze the log data using intelligent techniques. This project is focused on exploring the application of artificial intelligence (AI) in the incident response process to improve the speed, accuracy, and efficiency of the incident response process. The main objective of this project is to analyze the application of AI-assisted techniques for improving the incident response process using machine learning-based tools. In this project, the same cybersecurity scenario will be considered for analysis using the AI-assisted approach to analyze the authentication log files for identifying potential threats to the organization. The project also demonstrates how AI can be used to detect brute-force attacks by identifying patterns in the data, such as repeated failed login attempts followed by a successful login attempt. The AI-assisted approach to analyzing the data is more efficient in comparison to a human analyst because it can quickly interpret the data. This again emphasizes the potential benefits that can be achieved by implementing AI in the field of cybersecurity in terms of efficiency. However, the project also evaluates the limitations faced by AI-based techniques. This includes its dependency on data provided to it, the possibility that wrong data is provided to it, and its lack of understanding in a given context. While AI methods can assist in obtaining valuable data, they should not be used to replace human expertise. In essence, what this project has shown is that AI-based methods can play an important role in improving the efficiency of cybersecurity incident handling. This project can serve as a starting point to evaluate the potential benefits offered by AI-based cybersecurity techniques.

INTRODUCTION

The rapid development of digital systems and environments has also affected the quantity of cybersecurity threats that modern organizations should be able to deal with. Due to the complexity of modern cyberattacks, it has become essential for cybersecurity teams to process large quantities of data in order to deal with different security threats. Manual response to various security incidents, while essential, may be time-consuming. Manual response may not be adequate to handle the quantity of modern security operations (Kaushik, 2025).

In recent times, the role of artificial intelligence (AI) has become an essential part of modern cybersecurity. This is because AI offers the prospect of analyzing vast amounts of data, identifying patterns, and assisting organizations in various decision-making processes (Ban, 2023). This study examines the role of AI-based techniques in improving incident response within a simulated cybersecurity environment. This project builds on the concept of a simulated manual response to security incidents, as examined in Project 1. This study introduces an AI-based technique to analyze authentication logs to detect various security threats. The same attack scenario of repeated failed login attempts followed by a successful login attempt will be used to analyze the effectiveness of AI-based techniques compared to the manual response.

The primary aim of this project is to examine how AI technologies can help in enhancing the incident response process. By using AI to analyze the log data, the project aims to show how machine-based analysis can help in enhancing human expertise. The other aim of this project is to show how AI can help in enhancing cybersecurity. While it can help in enhancing cybersecurity through fast results and in identifying patterns, there is a possibility that it may not understand the context of the situation and may depend on the quality of the data. This project aims to show how the capabilities of AI need to be utilized in conjunction with human expertise to make the incident response process effective.

In conclusion, this project aims to show how AI can help in enhancing cybersecurity. By showing how AI can help in enhancing the incident response process, the project aims to show how AI can help in transforming the way in which cybersecurity is done. By showing how AI can help in enhancing the manual analysis process, the project aims to show how AI can help in enhancing the future of cybersecurity.

OBJECTIVES

The main objective of this project is to design and evaluate the application of an artificial intelligence-aided incident response methodology that can improve the analysis of cybersecurity incidents. The project is a continuation of a previously designed manual incident response simulation project. The main idea behind this project is to show how artificial intelligence can be applied to the incident response process to make it more efficient. One of the main objectives of this project is to apply an artificial intelligence-aided tool to analyze the authentication log data and determine the potential indicators of compromise. The project will provide the same log data that is being used for manual analysis to the artificial intelligence system to determine how effectively the artificial intelligence system can detect unusual patterns in the log data, determine the type of attack that is being carried out, and provide insights into the unusual behavior that is being detected.

Another main objective of this project is to compare the results of the artificial intelligence-aided analysis with those of the manual analysis. The aim of the project, therefore, will be to show that, although AI systems have the potential to speed up analysis, which will ease the cognitive burden of the analysts, there is a need for humans to interpret the results. Another aim of the project will be to examine the potential of AI systems in the triage of security incidents, especially with regard to prioritization and classification of security threats. In a real-world scenario, security analysts are often overwhelmed by the number of security threats, which makes it hard to identify which ones need urgent attention. The project, therefore, aims to demonstrate the potential that AI systems have in identifying security threats, particularly those that are deemed to be a major security threat, as well as identifying the most appropriate action to take in any situation.

Lastly, the project aims to demonstrate the importance that AI systems have in working in conjunction with other cybersecurity systems, as opposed to relying on AI systems alone to perform tasks that human analysts can perform. This, therefore, will align with contemporary cybersecurity systems, which have come to embrace a combination of human intelligence and AI systems. Generally speaking, the aim of this project is to make sure that there is a way to show that AI-aided techniques have been utilized within a controlled environment to simulate

conditions that are close to real-life. This way, it becomes possible to create an opportunity for further investigation into future AI-aided cybersecurity technologies.

BACKGROUND AND CONCEPTS

The complexity of cybersecurity threats and the data generated by modern technology have been some of the challenges affecting traditional cybersecurity processes. Due to the increased use of digital technology, the need for efficient cybersecurity solutions has been more critical. Artificial intelligence (AI) and machine learning (ML) have been some of the leading technologies that have been recognized as having a major influence on the transformation of cybersecurity solutions.

Artificial intelligence has been defined as the ability of computer systems to perform tasks considered intelligent for humans (Khandan et al., 2025). Artificial intelligence is normally used in cybersecurity to analyze data with the aim of identifying anomalies that may be indicative of cybersecurity threats. Machine learning, on the other hand, can be described as a type of artificial intelligence that helps computer systems to learn from the data, they receive to enhance their efficiency without necessarily being programmed to do so. Machine learning is usually applied to detect unknown cybersecurity threats.

One of the most important uses of AI technology in the field of cybersecurity is anomaly detection. Normally, cybersecurity uses specific sets of rules to help in the detection of attacks. Such systems may not be highly efficient in dealing with attacks that are unfamiliar. However, AI-based systems can help in the detection of anomalies that may be indicative of attacks (Wang, et al., 2024). This helps in the detection of attacks that may be subtle. Such attacks can also be sophisticated and not easily identifiable.

Another crucial concept that needs to be understood by a person who wishes to pursue a career in AI-based cybersecurity is incident triage. According to (Malve et al., 2025), Incident triage can be described as the process of prioritization and classification of security incidents depending on their severity. There may be numerous security incidents to deal with in a given situation. It may also be difficult to determine which ones are of critical importance. AI-based systems can help in the efficient handling of such situations by analyzing the data that comes into the system.

Additionally, another area where AI technology is frequently used is log analysis. Log analysis involves analyzing a number of system logs to determine patterns. This can be a tedious process, especially if the amount of data to be analyzed is significant. This process may also be prone to human error. The use of AI technology can help to hasten this process significantly. This will enable the human analyst to be more efficient while responding to an incident.

Although there are some advantages to the use of AI technology, there are some disadvantages to be considered. First, AI technology depends on the accuracy of the data it uses to analyze (Pallakonda et al., 2025). In case the data used to analyze an incident is false, it may result in false analysis. Another disadvantage of AI technology is that it does not have the power of understanding. This means that it may not be able to comprehend the context of the event. Therefore, it is best to use AI technology as an aid rather than relying on it.

Thus, the incorporation of AI in incident response is a move towards smarter and adaptive incident responses. It is believed that incorporating AI in incident responses is a best practice in modern cybersecurity. The concepts discussed in this section have offered a theoretical foundation to the AI-assisted incident response strategy employed in this project. The project has demonstrated how AI and machine learning can be leveraged to solve cybersecurity problems, thus enhancing the effectiveness of traditional incident responses.

IMPLEMENTATION

The focus of the implementation of the project is centered on the application of the AI-assisted approach in analyzing the authentication logs, as well as in assisting in incident response activities. Unlike the manual implementation, which relied on the direct analysis of the logs using command-line tools, the AI-assisted approach relies on the application of an artificial intelligence system in analyzing the same logs.

AI Tool Utilization

The AI-assisted approach in the implementation of the project utilized a natural language processing-based AI tool, which can analyze the input provided in a natural language and identify patterns in the input. The logs generated in the simulated environment, which contain information on the failed attempts at login followed by a successful login from the same source IP address, were provided as input to the AI tool in their raw form. The AI tool was then asked to

analyze the input logs and determine if there was a potential security threat in the activity being described in the logs. The input provided to the AI tool included raw logs, which contain information on the timestamp, username, source IP, and result of the login attempt.

AI Analysis Process

Upon receipt of this log data, the AI system carried out a pattern recognition process to detect unusual behavior within the authentication logs. The AI system detected repetitive attempts to log in to the system under a common username. The AI system also identified repetitive attempts made to log in to the system with a single IP address. The AI system identified repetitive attempts made to log in to the system, followed by a successful authentication attempt. This is a critical indicator of compromise. The AI system, based on this observation, concluded that this was a brute-force attack that had been successful. Apart from this conclusion, the AI system offered a detailed explanation of this attack. This involved determining the rationale behind this attack. This included determining what are some of the key indicators that this was a brute-force attack. These included repetitive authentication failures, rapid attempts to log in, and successful access from a similar source.

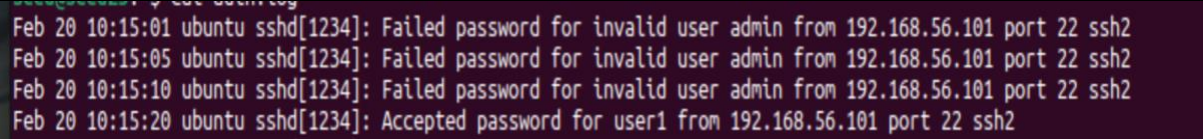
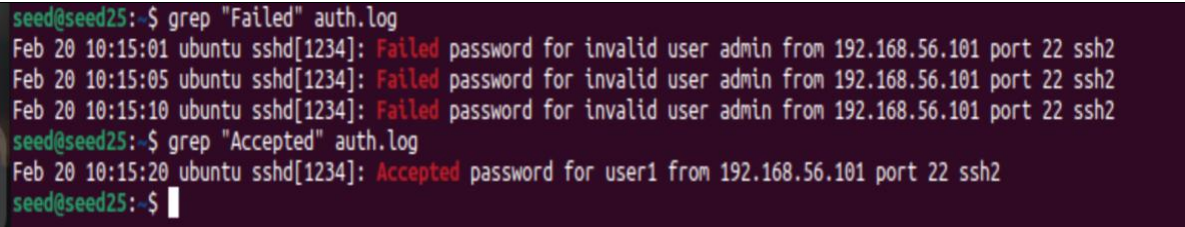
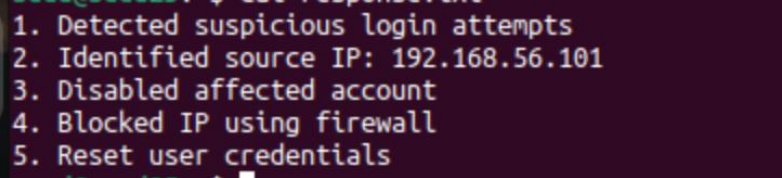
Response Recommendations

Other than the identification and categorization of the incident, the AI system offered recommendations on what to do to mitigate the identified threat. The recommendations included blocking the IP address, securing the user accounts, enhancing password security, and monitoring the activities to detect any suspicious behavior. The recommended actions provided by the AI system are similar to those provided in Project 1, which suggests that AI can be used in providing recommendations to aid in decision-making in incident handling based on data observations.

Implementation Outcome

The successful implementation of the technique with the help of AI was able to show the potential of an intelligent system that could analyze the log data, detect abnormal patterns, and classify cybersecurity incidents with minimal human intervention. The AI system was able to generate accurate interpretations of the log data, which helped minimize the time taken to analyze the data.

In addition to this, the output generated by the AI system was clear and concise in nature. This would help in easily understanding the nature of the attack and the appropriate actions to be taken in response to the attack. Thus, the successful implementation of the AI-assisted technique in cybersecurity confirms that such techniques can be used to effectively implement the manual processes in handling cybersecurity incidents. This would help in ensuring the efficiency and scalability of the AI system in handling such incidents.

INCIDENT DETECTION	
	
Figure 1	
Authentication logs showing repeated failed login attempts followed by a successful login	
	
Figure 2	
Filtered log output using grep command to identify suspicious activity	
	
Figure 3	
Documented incident response actions	

What the log shows

1. Multiple failed login attempts

code

 Copy

```
Failed password for invalid user admin from 192.168.56.101
(repeated several times)
```

Key observations:

- Repeated attempts
- Same username (admin)
- Same source IP (192.168.56.101)
- Over a short time interval

2. Then a successful login

code

 Copy

```
Accepted password for user1 from 192.168.56.101
```

This is the red flag moment:

- Same attacking IP
- Now successfully logged in
- Different username (user1)

Attack Classification

Brute Force Attack (SSH Brute Force / Password Guessing)

More specifically:

- |  Online brute-force attack leading to account compromise

Attack Type:

- | SSH Brute Force Attack (Password Guessing)

Impact Level:

- | High – resulted in successful account compromise

Evidence:

- Multiple failed login attempts from same IP
- Targeting common usernames
- Followed by a successful login from same IP

Figure 4

AI-assisted analysis of authentication logs

As shown in Figure 1, the authentication logs show repeated failed login attempts by one IP address. The repeated failed login attempts show abnormal patterns. This is an example of how the analysis may be utilized to identify indicators of potential attacks.

Further analysis, as shown in Figure 2, is done by using command-line filtering. This again identifies the repeated failed login attempts, showing abnormal patterns. This is an example of how manual analysis may be utilized to identify indicators of potential attacks.

The response actions, as shown in Figure 3, show the steps that were undertaken to mitigate the identified threat. The steps include blocking the IP and securing the accounts.

Finally, Figure 4 shows the AI-assisted analysis. This analysis identifies the same patterns in the logs and has identified the attack. The AI has identified the type of attack and has given a structured analysis. The AI has also given the recommended response actions. This is an example of how AI may be utilized to enhance the incident detection process.

RESULTS AND DEMONSTRATION

The results of this project clearly prove the effectiveness of using artificial intelligence in the incident response process. The AI system, using the same authentication log information obtained in the manual incident response simulation, successfully identified suspicious activity, classified the incident as a specific type of cyber-attack, and provided a list of response actions needed in a minimal human intervention fashion.

The AI system was able to analyze the authentication log, identify a pattern in the activity which seemed suspicious, and finally classify it as a brute-force cyber-attack. The AI system identified a number of failed login attempts coming from a single IP address. The AI system realized that the failed login attempts were carried out within a short period of time and targeted common username formats, which is a common characteristic of a brute force cyber-attack. The AI system also detected a successful login attempt from the same IP address; the AI system realized that it is a successful brute force cyber-attack.

The AI system successfully classified the activity as a brute force cyber-attack through its analysis of the activity; this is in line with what was derived from the manual analysis in Project 1. This further proves that AI can properly interpret the data in the logs and can identify common

patterns of attacks with enough contextual information provided. One of the most important findings in working with AI is its ability to perform an analysis in a very efficient manner. Although it took a lot of time to manually analyze and observe patterns in the logs, the AI had achieved its task in an instant. This is perhaps one of the most important contributions AI could make to the security system.

Moreover, the AI had offered a structured explanation of the observed behavior, which helped in understanding the nature of the attack. The AI had offered the response actions, which included blocking the IP address, securing the user account, etc. These response actions were also similar to those identified in the manual implementation, which further proves the effectiveness of using AI in cybersecurity.

The results clearly indicate that AI can be a powerful tool in cybersecurity, which can be used in the analysis of log data and can help in decision-making. However, it is also clear from the results that AI can be an effective tool only in conjunction with human expertise. Although it is evident from the result that the nature of the attack is being correctly identified by the AI system, there is also a need for human expertise to verify the result. The demonstration has clearly shown that the effectiveness of using AI in cybersecurity is evident, which is helpful in improving the efficiency of cybersecurity analysis. There is also a possibility that using AI along with human expertise will provide a more effective way of dealing with cybersecurity.

DISCUSSION AND ANALYSIS

The results obtained from this project have shown that the application of artificial intelligence within the incident response process could greatly improve efficiency and effectiveness within the field of cybersecurity analysis. As demonstrated within this project; by applying an AI-assisted approach to the same dataset as that used within the manual simulation, it has been possible to observe notable differences within speed, scalability, and interpretation ease between the two methods. The most notable advantage that has been demonstrated within this project is that of speed. As demonstrated within this research; by applying an AI-assisted approach to the analysis of authentication logs, it has been possible to achieve an almost instantaneous interpretation, as opposed to a more time-consuming approach that required recognition on the part of the individual. This is particularly important within a real-world environment, as there are often vast amounts of log data that need to be constantly analyzed. This is where AI could greatly

assist within the field of cybersecurity analysis, as it could greatly reduce time and allow for faster identification of security threats.

Another major advantage that this AI-assisted approach has over the traditional approach is that, with this approach, a structured output is possible. Not only did the AI system find the pattern, but it also gave a proper explanation of its thought process. This will be helpful to inexperienced analysts because they will be able to understand the process. This will prevent them from getting confused. On the other hand, this approach is totally dependent on the skills and expertise of the analyst. There are certain limitations of using artificial intelligence as a tool for incident response as well. The first major limitation of using AI for incident response is its inability to understand the context. Although the AI was able to identify the type of attack, there might be situations when the context will need to be considered as well. For instance, differentiating between normal system usage and malicious activities might not be easy without proper context.

The next limitation is that AI can be based on assumptions that are not correct. This is because AI can provide important information, but it is not always correct. Thus, it shows that AI can come to the wrong conclusion based on the wrong data. This again emphasizes the importance of human intervention in AI so that it does not come to the wrong conclusion. Cybersecurity analysts should be involved in AI so that it does not come to the wrong conclusion.

Lastly, it is evident that AI technology depends on data input. Therefore, if the data that goes into AI technology is wrong or badly structured, it may not work properly. In this project, data was properly defined. Therefore, AI worked properly to give results. However, in a real-world situation, it may be badly defined. This may be a challenge to AI technology.

Despite these limitations, the project proves that AI-assisted incident response has significant benefits when used in conjunction with human analysis. This is because the use of human expertise in conjunction with AI creates a more efficient model in cybersecurity. This is because AI has the ability to process data as well as make initial decisions. In conclusion, this discussion has proven the need to use a hybrid approach when dealing with cybersecurity attacks. This is because AI should not be used to replace human expertise when dealing with cybersecurity attacks. Rather, AI should be taken as a complementary approach that works hand in hand with human expertise. This will help organizations respond to cybersecurity attacks efficiently.

FUTURE WORK

Although this project shows the efficiency of using AI tools to analyze and classify cybersecurity threats, there are various opportunities that could be added to make this project even better, thereby enhancing efficiency within a real-world environment. The first opportunity to make this project better, thereby enhancing efficiency, is to incorporate more efficient machine learning algorithms, especially those that have been trained on cybersecurity data. In this project, a general AI tool was employed to analyze the data. Nevertheless, more sophisticated tools, especially those that have been trained on large datasets, could improve the efficiency of the system. This could be achieved through supervised learning algorithms.

Another area where this project may be enhanced is in the incorporation of real-time processing and monitoring. Currently, the analysis of AI takes place on static logs after the occurrence of the event. Real-time analysis, however, may be necessary in real-world security operations to effectively detect and respond to emerging threats. This project, therefore, may be enhanced by the inclusion of AI with real-time log analysis to facilitate real-time monitoring and threat detection.

Another area that this project may be enhanced to cover is the inclusion of AI with Security Information and Event Management (SIEM) systems. This is because SIEM systems are normally used in enterprise organizations. The inclusion of AI with SIEM systems will help organizations to enhance their capabilities to correlate event data from various systems, thus gaining a more comprehensive understanding of the security incident. Another area that can also be considered in future development is the explainability of AI. This is significant in the sense that although AI may help one understand the situation to a limited extent, it is also crucial that the analyst understands how the decision was made. Therefore, future development can also aim at creating an explainable AI that can help one understand the results.

Additionally, an area that could be considered for future development of this project is the use of AI against more sophisticated attacks. This could include multi-stage attacks, lateral movement attacks, and even advanced persistent threats. This will provide a better understanding of what AI can achieve in the incident response process. Lastly, this project can provide a basis for further development in fully automated incident response systems that utilize AI in both the detection

and response actions. While this in itself creates problems and risks, it does open up a potential avenue to improve response times and reduce human involvement in large-scale systems.

In conclusion, further developments in advanced machine learning, real-time analysis, system integration, and explanation have the potential to greatly increase the effectiveness of AI in incident response systems. This is in line with the changing environment in cybersecurity and can provide a basis for further developing more intelligent systems in cybersecurity.

CONCLUSION

This project has been a successful example of the integration of artificial intelligence with the incident response process, which has highlighted the potential of artificial intelligence in enhancing the analysis of cybersecurity incidents. The project has been able to leverage a manual simulation of the incident response process, which has allowed a direct comparison to be made with the analysis of cybersecurity incidents with the aid of artificial intelligence and conventional analysis, with the same authentication log information being used to ensure accuracy.

Based on the analysis, it has been quite evident that the artificial intelligence has the potential to identify patterns in the log data, identify indicators of compromise, and even classify security incidents with a high level of accuracy. In this particular analysis, the artificial intelligence has been able to identify a sequence of failed login attempts followed by a successful authentication, which is a brute force attack resulting in unauthorized access, just like in the manual analysis.

One of the most important advantages that have been observed with this particular project is the efficiency of the AI system, which has allowed the analysis to be conducted in a swift manner. The AI system has been able to analyze the data almost instantly, which is a major advantage, considering the dynamics in the field of cybersecurity, which has a lot to do with analyzing the data at a swift pace.

Furthermore, as seen in this particular project, there is a need for the AI system not to replace human analysis but to provide a means of support. This is because there are instances where the AI system does not understand the context of the data; hence, there is a need for human analysis to ascertain that the data has been analyzed correctly.

The application of this combination of analysis techniques is seen to provide a hybrid solution that takes advantage of the benefits of each of these techniques. The application of the AI-based techniques in the incident response process of an organization will help to promote the efficiency of the overall incident response of the organization to cybersecurity threats while at the same time maintaining the significance of human analysis. In conclusion, this project proves that AI-assisted incident response is an important development in the field of cybersecurity. This project provides an avenue for further exploration of the idea of intelligent security systems.

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